Literature Answer: Davis Endpoint for Noncommutative Symmetric Spaces

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2026-07-18

Status. `literature_already_answered`. The extracted endpoint Davis problem from arXiv:1501.05944 is answered in later literature.

Source question. Randrianantoanina–Wu, *Martingale inequalities in noncommutative symmetric spaces*, arXiv:1501.05944, asks in the final section whether the assumption $p_E > 1$ can be removed from the first equality in their Davis corollary. In their notation:

$$p_E = 1, \quad q_E \leq 2 \qquad \Longrightarrow \qquad \mathcal{H}_E(\mathcal{M}) = \mathsf{h}_E(\mathcal{M})\ ?$$

Later answer. Randrianantoanina, *P. Jones' interpolation theorem for noncommutative martingale Hardy spaces*, arXiv:2212.08714, proves in Proposition `Davis-symmetric` that if $E \in \mathrm{Int}(L_1, L_2)$, then the adapted column sequence space decomposes as

$$E^{\mathrm{ad}}(\mathcal{M}; \ell_2^c) = E\left(\bigoplus_{n\geq 1} \mathcal{M}_n\right) + E^{\mathrm{cond,ad}}(\mathcal{M}; \ell_2^c).$$

Its Corollary `Davis` gives the martingale formulation

$$\mathcal{H}_E^c(\mathcal{M}) = \mathsf{h}_E^d(\mathcal{M}) + \mathsf{h}_E^c(\mathcal{M}),$$

and hence, by the row version,

$$\mathcal{H}_E(\mathcal{M}) = \mathsf{h}_E(\mathcal{M}).$$

The text immediately before the corollary says this answers [1, Problem 4.2], where [1] is exactly the arXiv:1501.05944 paper.

Sharp threshold. The same corollary records sharpness: if the column Davis identity holds, then already $\mathsf{h}_E^c(\mathcal{M}) \subseteq \mathcal{H}_E^c(\mathcal{M})$, and a two-step commutative martingale argument plus Kalton–Montgomery-Smith's criterion forces

$$E \in \mathrm{Int}(L_1, L_2).$$

Thus the correct endpoint criterion is $E \in \mathrm{Int}(L_1, L_2)$, not merely $p_E = 1$ and $q_E \leq 2$.

Consequence for the 2015 Boyd-index formulation. If $q_E < 2$, standard Boyd interpolation implies $E \in \mathrm{Int}(L_1, L_2)$, so the answer is affirmative. At the boundary $q_E = 2$, the literal formulation is false in general. For example, $E = L_1 + L_{2,\infty}$ is a symmetric Fatou Banach function space with Boyd indices $p_E = 1$ and $q_E = 2$, but it is not an interpolation space for (L_1, L_2): weak-L_2 contains functions not in $L_1 + L_2$. The sharpness result therefore rules out the Davis equality for this E in general.

Why this is not new. The supporting paper explicitly identifies the original Randrianantoanina–Wu problem and gives both the positive endpoint theorem and the sharpness criterion.

References

[1] N. Randrianantoanina and L. Wu, *Martingale inequalities in noncommutative symmetric spaces*, arXiv:1501.05944.

[2] N. Randrianantoanina, *P. Jones' interpolation theorem for noncommutative martingale Hardy spaces*, arXiv:2212.08714.

[3] N. Kalton and S. Montgomery-Smith, *Interpolation of Banach spaces*, Handbook of the Geometry of Banach Spaces, Vol. 2, North-Holland, 2003.